

Does Physical Exhaustion Level Affect Memory?

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Our Goal

Research Questions:

- What are the effects of rest and exercise on memory?
- Does sleep significantly boost memory?
- What level of exercise is needed to see an effect on memory?
- Does the impact of exercise on memory depend on how much sleep you get?



Literature Review

Memory can be simply defined as the ability to recall information. Memories are responsible for the stories we tell ourselves and help build the basis of our relationships. Losing your skill to maintain a memory hurts basic yet essential daily tasks.

Many academic sources have investigated factor that may influence memory. Two typically factor seen to help memory in various different aspects is sleep and exercise.

- Brain receives a boost after physical activity within the first 10 to 20 minutes (Medical News Today).
- Physical exercise improves declarative memory and procedural memory (Roig, 2022).
- Research by Kuriyama shows higher working memory accuracy after sleep.
- People who exercise and nap show greater recall abilities more than people who either exercised or took a nap only (Roig, 2022)

Design

Randomized Complete Block Design with 2x2 Factorial

- **Factor A - Rest**
 - 2 levels: Nap 15 minutes and Sit 10 minutes
- **Factor B - Exercise**
 - 2 levels: Run Indoors 1 km and Run Indoors for 100 m
- **Block - Age**
 - 5 age groups: 18-25 year olds, 26-35 year olds, 36-45 year olds, 46-55 year olds, 56-65 year olds
- **Interaction between Rest and Exercise**
 - 4 groups in each block
- **Response - Memory**
 - Memory Game
 - Time (in seconds) taken to match 30 pairs of cards



Response	(time in seconds)	Memory Game
Predictors	Run Type	Run Indoors 1km, Run Indoors 100m
	Rest Type	Nap 15 minutes, Sit 10 minutes
Block	Age	18-25, 26-35, 36-45, 46-55, 56-65



Benchmark
df = 1



Rest
df = 1



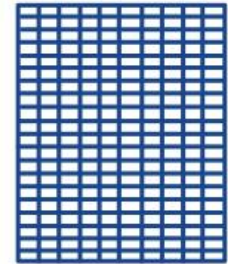
Exercise
df = 1



Interaction
df = 1



Block (Age)
df = 4

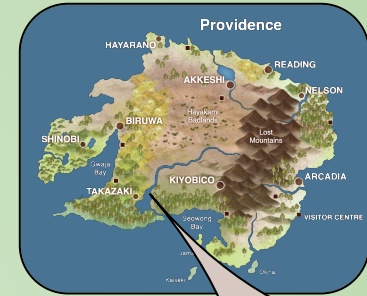


Error
df = 192

Sampling Method

How we sampled:

- Multistage sampling with three stages - used a random number generator to (pseudo) randomly select an item at each stage
- First stage: randomly select a town on Providence Island
- Second stage: randomly select a house within that town
- Third stage: randomly select a person within that house
- Repeated for all 200 subjects

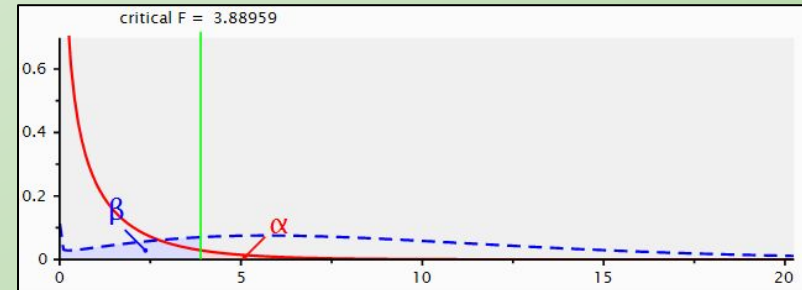


House 63

Aadiv Laha	36
Justine Rose	36
Rohan Laha	16
Alexis Rose	14

Sample size determination:

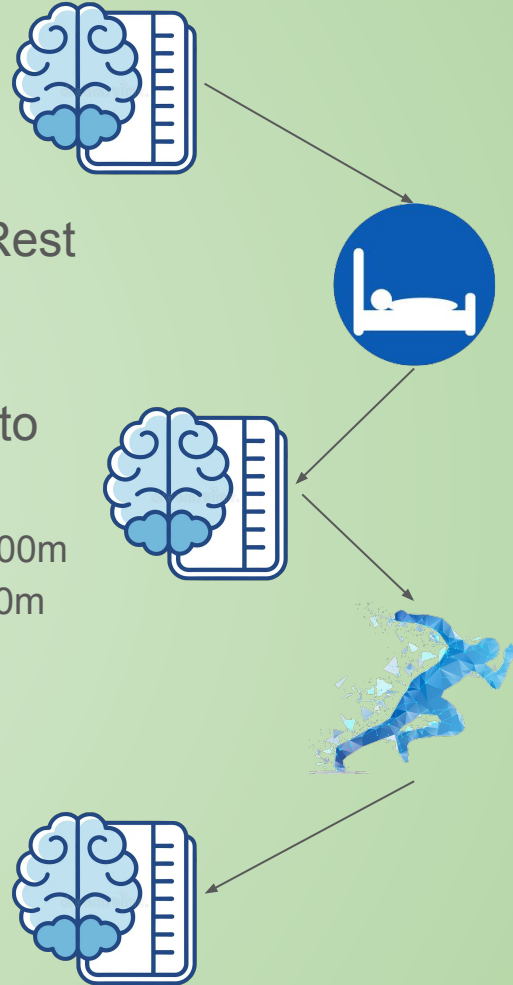
- Power - 0.8
- Alpha - 0.05
- Effect Size - 0.2
- G*Power Recommended Sample Size - 198
 - Therefore each age group had 40 subjects (rounded up)



Procedure

5 blocks by age group: 40 islanders in each block

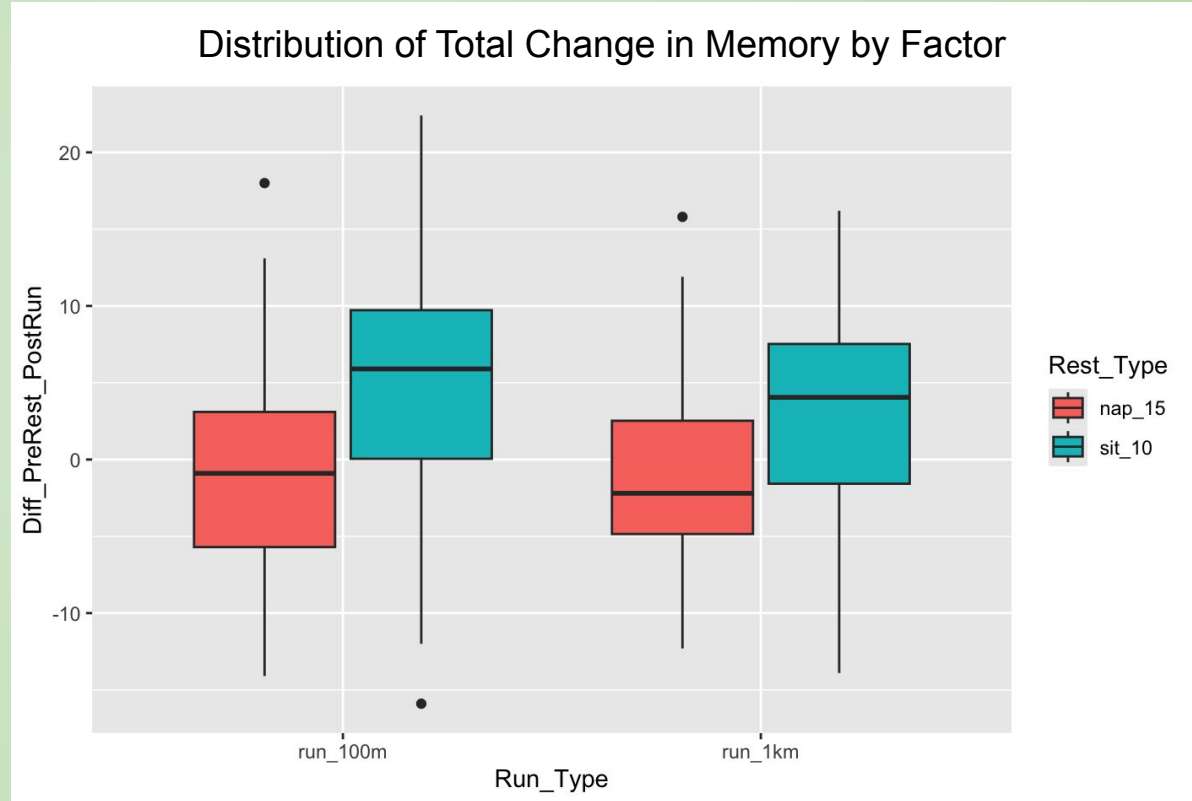
- Each age group is randomly assigned into each of the two Rest Conditions:
 - 20 islanders in the Nap condition, 20 islanders in the Sit condition
- Within each Rest group, islanders are randomly assigned into each of the Run Conditions:
 - NAP: 10 islanders Run Indoors for 1km, 10 islanders Run Indoors for 100m
 - SIT: 10 islanders Run Indoors for 1km, 10 islanders Run Indoors for 100m
-
- Timed during a Memory game for a baseline measurement
 - Complete REST condition
 - Timed during Memory game
 - Complete EXERCISE condition
 - Timed during Memory game



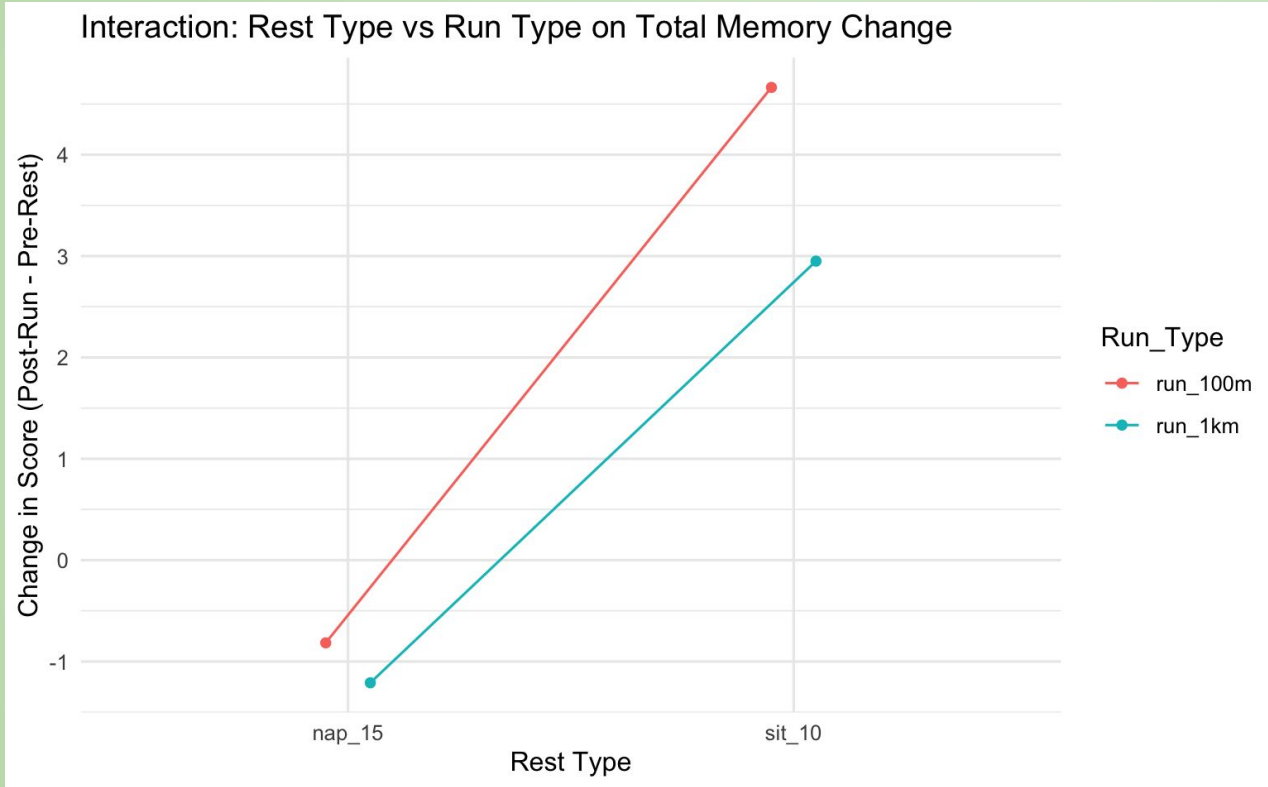
Exploratory Analysis

Looking at the distributions of each group, we see that the two nap groups are centered around negative values.

This indicates that napping may have a positive effect on memory, but further investigation is required to be sure.



Is there an Interaction?



The interaction plot between Rest Type and Run Type (after resting) suggests a there is likely no interaction effect, indicating that the combination of rest method and running distance may influence overall memory performance.

This implies that how someone rests and the intensity of subsequent exercise together don't impact memory retention. We will explore this interaction in more detail in the coming slides.

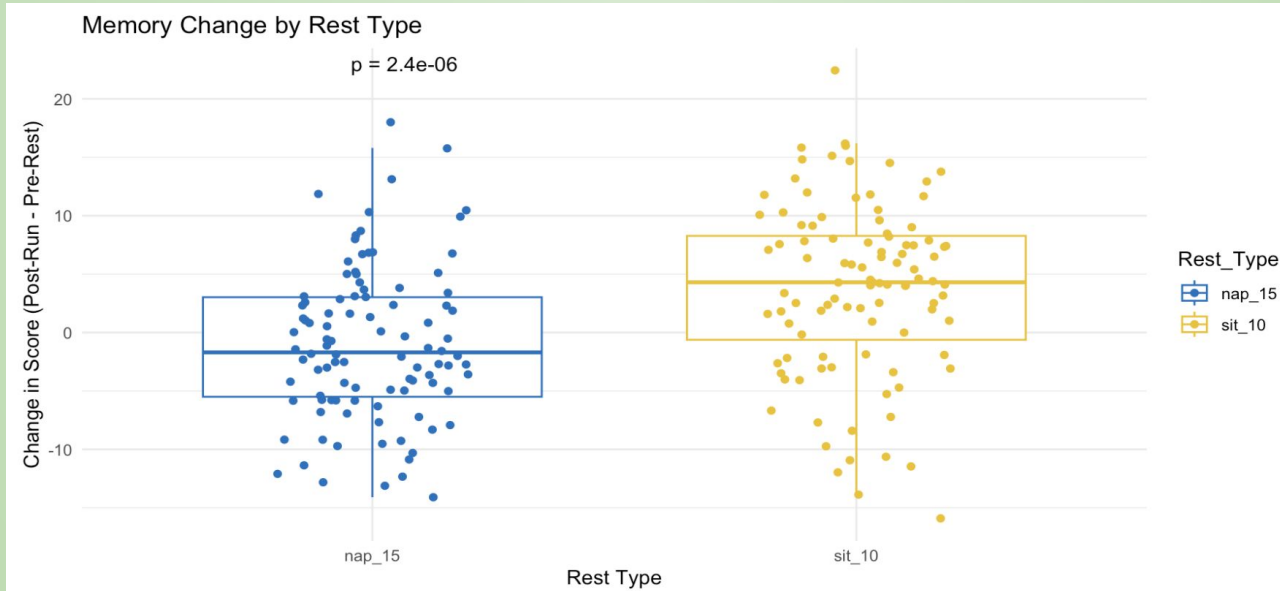
ANOVA Plot

The ANOVA results below show us that after controlling for age-related variability, Rest_Type still has a clear and strong effect on the performance outcome. Run_Type does not. There's also no evidence that Rest_Type's effect depends on Run_Type confirming what the previous graph suggested, the p-value of the interaction ($p = 0.5003$) is much greater than our threshold ($p = 0.05$)

	DF	Sum Sq	Mean Sq	F-Value	Pr(>F)
Rest_Type	1	1162	1161.6	24.327	1.76e-06
Run_Type	1	56	55.5	1.163	0.2821
Age_Group	4	502	125.4	2.627	0.0359
Rest_Type:Run_Type	1	22	21.8	0.456	0.5003
Residuals	192	9168	47.8		

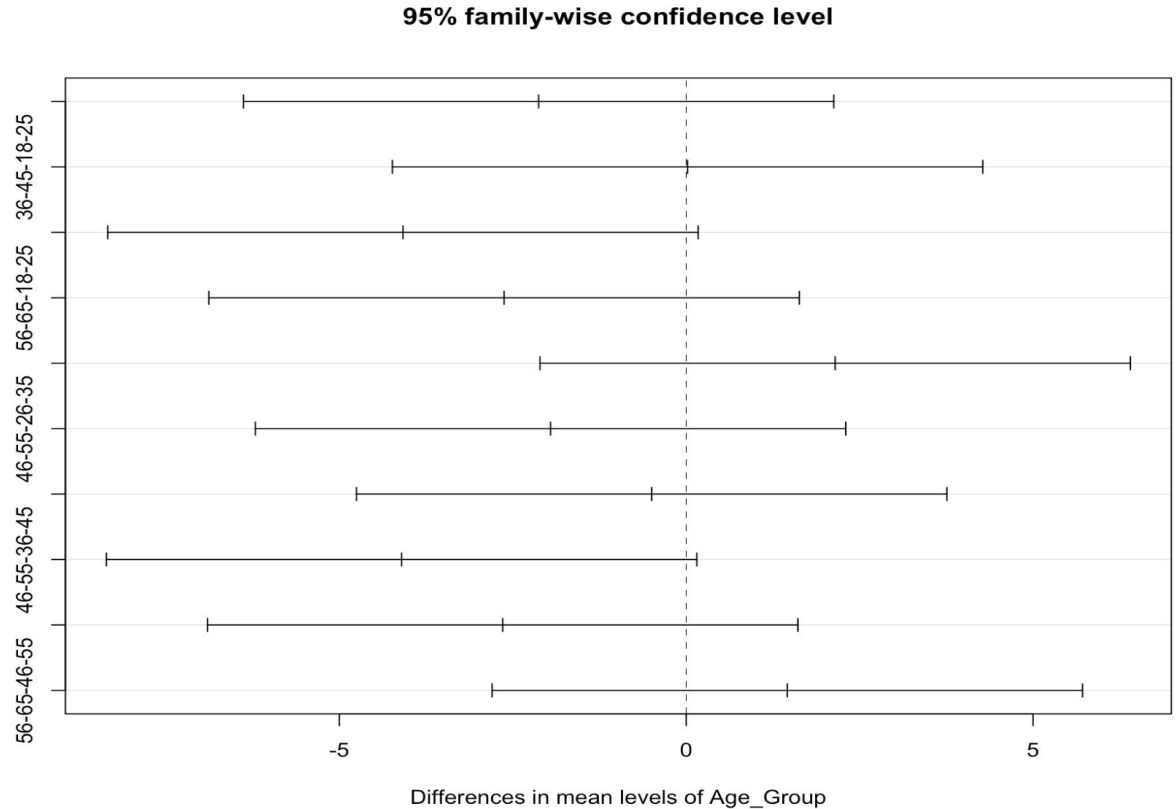
T-Test

From the plot, the nap 15 group has a lesser mean than the sit 10 group. This indicates that participants who napped for 15 minutes improved their memory more compared to those who just sat quietly. The confidence intervals for the two groups are well-separated, and the t-test confirms that this difference is statistically significant. Therefore, we conclude that napping led to significantly greater improvements in memory performance (as measured by time reduction) than sitting quietly.

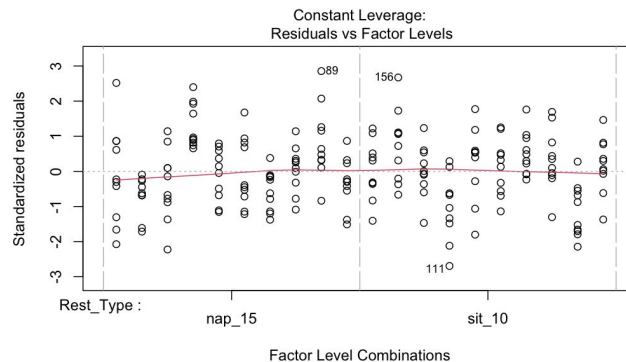
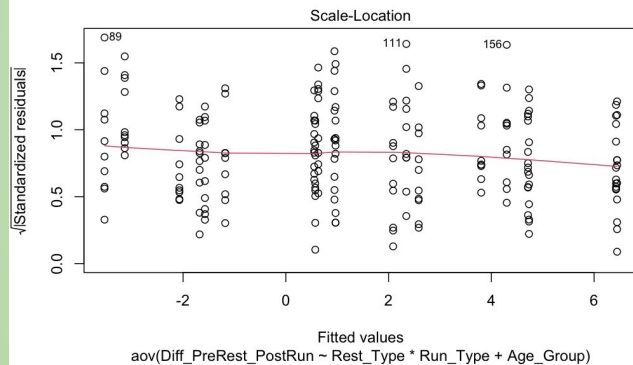
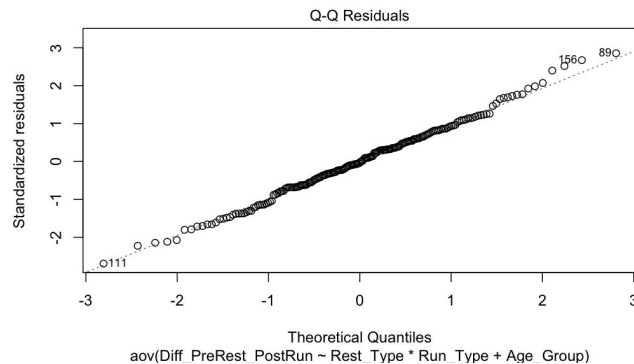
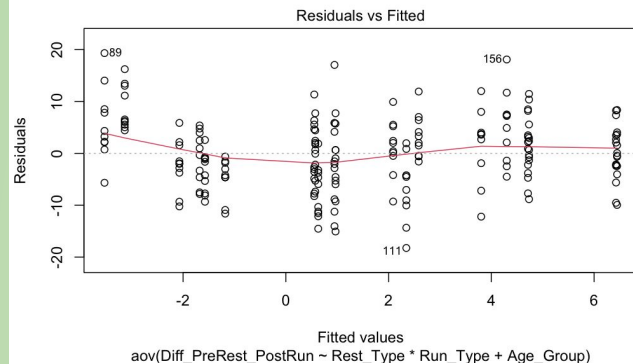


Tukey Plot - Age Groups

There are no statistically significant pairwise differences between individual age groups, even though the overall ANOVA showed that Age_Group has a significant effect.



Residual Plots & Model Assumptions



The assumptions of the model are met making the model a valid model and there for the conclusions are valid as there are no violations in assumptions.

Summary

Our Goal was to examine the impact of different rest methods and exercise intensities affect memory performances.

- **Rest Type significantly impacts memory:** ANOVA showed a very low p-value for rest type, indicating strong evidence that how a person rests influences memory
- **Running distance had no significant effect:** Memory performance did not differ meaningfully between the 1km and 100m run groups
- **No interaction effect:** Rest method and run intensity do not interact – their combination does not have a unique impact on memory beyond their individual effects

Resting method plays a more critical role in memory retention than than physical activity.



Future Research Questions

- Are boosted memory effects of exercise and rest seen other types of memory? (ex. procedural and semantic memory)
- Do different stages of sleep (ex. short nap vs REM sleep) impact physical activity and memory?
- How does regular exercise affect long-term memory function?



References

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- Sleep Accelerates the Improvement in Working Memory Performance
 - <https://www.jneurosci.org/content/28/40/10145>

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- Exercise and deep sleep give the brain a 24-hour boost
 - <https://www.medicalnewstoday.com/articles/exercise-and-deep-sleep-give-the-brain-a-24-hour-boost>